ANSWER KEY & MARKING SCHEME — SET 1

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{25}{9}$	$1+\tan^2\theta$
2	**(a)** $-\frac{7}{25}$	$\cos^2\theta-\sin^2\theta$
3	**(c)** $\frac{24}{25}$	$2\cdot\frac{p}{c}\cdot\frac{q}{c}$
4	**(a)** $\frac{1}{3}$	$(\sec+\tan)(\sec-\tan)=1$
5	**(b)** 1	Each pair sums to 1
6	**(a)** 1	Pair as $\tan\theta\tan(90^\circ-\theta)=1$
7	**(b)** $\frac{169}{25}$	$1+\tan^2\theta$
8	**(d)** $-\frac{119}{169}$	$\cos^2\theta-\sin^2\theta$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{3}{4}+\frac{1}{4}=1$; A is a particular case of the identity in R.
10	**(c)** A is true but R is false.	$\tan 45^\circ=\frac{1/\sqrt{2}}{1/\sqrt{2}}=1$: A true; $\tan\theta=\frac{\sin\theta}{\cos\theta}$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\cos^2\theta=1-\left(\frac{4}{5}\right)^2=\frac{9}{25}$ (1); value $=-\frac{7}{25}$ (1).	1+1
12	Hypotenuse $=29$ (1); $\sin\theta=\frac{21}{29}$, $\sec\theta=\frac{29}{20}$ (1).	1+1
13	$\frac{x^2}{4}=\sec^2\theta$, $\frac{y^2}{9}=\tan^2\theta$ (1); difference $=1$ (1).	1+1
14	$\cos^2\theta=1-\left(\frac{12}{13}\right)^2=\frac{25}{169}$ (1); value $=-\frac{119}{169}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LHS $=\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}$ (2); $=\frac{1}{\sin\theta\cos\theta}=\sec\theta\csc\theta$ (1).	2+1
16	Square both sides: $1+2\sin\theta\cos\theta=\frac{49}{25}$ (2); $\sin\theta\cos\theta=\frac{12}{25}$ (1).	2+1
17	Square: $\tan^2\theta+2+\cot^2\theta=9$ (2); value $=7$ (1).	2+1
	OR: LHS $=\frac{1}{\cos^2\theta}+\frac{1}{\sin^2\theta}=\frac{\sin^2\theta+\cos^2\theta}{\sin^2\theta\cos^2\theta}$ (2); $=\sec^2\theta\csc^2\theta$ (1).	2+1
18	Square both sides: $1+2\sin\theta\cos\theta=\frac{169}{100}$ (2); $\sin\theta\cos\theta=\frac{69}{200}$ (1).	2+1
	OR: Square: $\tan^2\theta+2+\cot^2\theta=16$ (2); value $=14$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sin P=\frac{QR}{PR}=\frac{12}{13}$ (1). (ii) $\cos R=\frac{QR}{PR}=\frac{12}{13}$ (1). (iii) $\tan P=\frac{QR}{PQ}=\frac{12}{5}$ (1). (iv) $\sec P=\frac{13}{5}$; LHS $=1+\frac{144}{25}=\frac{169}{25}=\left(\frac{13}{5}\right)^2=$ RHS (2).	1+1+1+2
20	LHS $=\frac{\sin\theta(1-2\sin^2\theta)}{\cos\theta(2\cos^2\theta-1)}$ (2); $1-2\sin^2\theta=2\cos^2\theta-1$ (2); so LHS $=\tan\theta$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{3\tan A+7}{2\tan A+1}$ (1); $=\frac{3(\frac{21}{20})+7}{2(\frac{21}{20})+1}=\frac{203}{62}$ (2); verified with $\sin A=\frac{21}{29}$, $\cos A=\frac{20}{29}$ (1).	1+1+2+1

Mistakes to avoid:

- Treating $\sin A$ as $\sin\times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta=\frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.

- Writing $\sin 2A = 2\sin A$ or $\sin(A+B) = \sin A + \sin B$, which are false.
- Wrong table values, e.g. $\sin 30^\circ = \dfrac{\sqrt{3}}{2}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1	Each pair sums to 1
2	**(c)** 1	Pair as $\tan\theta\tan(90^\circ-\theta)=1$
3	**(a)** $\frac{289}{64}$	$1+\tan^2\theta$
4	**(a)** $-\frac{161}{289}$	$\cos^2\theta-\sin^2\theta$
5	**(d)** $\frac{240}{289}$	$2\cdot\frac{p}{c}\cdot\frac{q}{c}$
6	**(b)** $\frac{1}{4}$	$(\sec+\tan)(\sec-\tan)=1$
7	**(d)** 1	Each pair sums to 1
8	**(d)** 1	Pair as $\tan\theta\tan(90^\circ-\theta)=1$
9	**(d)** A is false but R is true.	$\sin\theta$ never exceeds 1, so A is false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both equal $\frac{1}{2}$ (A true) and R is also true, but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hypotenuse $=41$ (1); $\sin\theta=\frac{40}{41}$, $\sec\theta=\frac{41}{9}$ (1).	1+1
12	$\frac{x^2}{9}=\sec^2\theta$, $\frac{y^2}{25}=\tan^2\theta$ (1); difference $=1$ (1).	1+1
13	$\cos^2\theta=1-\left(\frac{15}{17}\right)^2=\frac{64}{289}$ (1); value $=-\frac{161}{289}$ (1).	1+1
14	Hypotenuse $=10$ (1); $\sin\theta=\frac{4}{5}$, $\sec\theta=\frac{5}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$1+\tan^2A=\sec^2A$ (1); LHS $=(1-\tan^2A)\cos^2A$ (1); $=\cos^2A-\sin^2A$ (1).	1+1+1
16	Square both sides: $1+2\sin\theta\cos\theta=\frac{289}{169}$ (2); $\sin\theta\cos\theta=\frac{60}{169}$ (1).	2+1
17	Square: $\tan^2\theta+2+\cot^2\theta=25$ (2); value $=23$ (1).	2+1
	OR: $(1+\tan^2A)=\sec^2A$ (1); $\sec^2A\sin^2A=\dfrac{\sin^2A}{\cos^2A}$ (1) $=\tan^2A$ (1).	1+1+1
18	Square both sides: $1+2\sin\theta\cos\theta=\frac{25}{16}$ (2); $\sin\theta\cos\theta=\frac{9}{32}$ (1).	2+1
	OR: Square: $\tan^2\theta+2+\cot^2\theta=36$ (2); value $=34$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{2}+\frac{1}{2}=1$ (1). (ii) $\sqrt{3}\times\frac{1}{\sqrt{3}}=1$ (1). (iii) $\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}-\frac{1}{2}\cdot\frac{1}{2}=\frac{3}{4}-\frac{1}{4}=\frac{1}{2}$ (1). (iv) $A+B=90^\circ$ and $A-B=30^\circ$ (1), so $A=60^\circ$, $B=30^\circ$ (1).	1+1+1+2
20	Use $1=\sec^2\theta-\tan^2\theta$ in the numerator: $(\sec\theta+\tan\theta)-(\sec\theta-\tan\theta)(\sec\theta+\tan\theta)$ (2); $=(\sec\theta+\tan\theta)(1-\sec\theta+\tan\theta)$ (2); cancel with denominator: $\sec\theta+\tan\theta=\dfrac{1+\sin\theta}{\cos\theta}$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\dfrac{2\tan A+3}{8\tan A+1}$ (1); $=\dfrac{2(\frac{4}{3})+3}{8(\frac{4}{3})+1}=\frac{17}{35}$ (2); verified with $\sin A=\frac{4}{5}$, $\cos A=\frac{3}{5}$ (1).	1+1+2+1

Mistakes to avoid:

- Treating $\sin A$ as $\sin\times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta=\dfrac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A=2\sin A$ or $\sin(A+B)=\sin A+\sin B$, which are false.

- Wrong table values, e.g. $\sin 30^\circ = \frac{\sqrt{3}}{2}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{336}{625}$	$2 \cdot \frac{p}{c} \cdot \frac{q}{c}$
2	**(c)** $\frac{1}{7}$	$(\sec + \tan)(\sec - \tan) = 1$
3	**(a)** 1	Each pair sums to 1
4	**(a)** 1	Pair as $\tan\theta \tan(90^\circ - \theta) = 1$
5	**(a)** $\frac{841}{400}$	$1 + \tan^2\theta$
6	**(a)** $-\frac{41}{841}$	$\cos^2\theta - \sin^2\theta$
7	**(d)** $\frac{840}{841}$	$2 \cdot \frac{p}{c} \cdot \frac{q}{c}$
8	**(d)** $\frac{2}{5}$	$(\sec + \tan)(\sec - \tan) = 1$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\cos A = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$, using the identity in R.
10	**(c)** A is true but R is false.	A is a standard identity (true). The correct identity is $1 + \cot^2\theta = \operatorname{cosec}^2\theta$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{x^2}{16} = \sec^2\theta$, $\frac{y^2}{49} = \tan^2\theta$ (1); difference $= 1$ (1).	1+1
12	$\cos^2\theta = 1 - \left(\frac{24}{25}\right)^2 = \frac{49}{625}$ (1); value $= -\frac{527}{625}$ (1).	1+1
13	Hypotenuse $= 37$ (1); $\sin\theta = \frac{35}{37}$, $\sec\theta = \frac{37}{12}$ (1).	1+1
14	$\frac{x^2}{25} = \sec^2\theta$, $\frac{y^2}{4} = \tan^2\theta$ (1); difference $= 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{\sin A}{\csc A} = \sin^2 A$, $\frac{\cos A}{\sec A} = \cos^2 A$ (2); sum $= 1$ (1).	2+1
16	Square both sides: $1 + 2\sin\theta\cos\theta = \frac{81}{49}$ (2); $\sin\theta\cos\theta = \frac{16}{49}$ (1).	2+1
17	Square: $\tan^2\theta + 2 + \cot^2\theta = 49$ (2); value $= 47$ (1).	2+1
	OR: $\sin^2 A \cdot \frac{\cos^2 A}{\sin^2 A} = \cos^2 A$ (1); $\cos^2 A \tan^2 A = \sin^2 A$ (1); sum $= 1$ (1).	1+1+1
18	Square both sides: $1 + 2\sin\theta\cos\theta = \frac{121}{64}$ (2); $\sin\theta\cos\theta = \frac{57}{128}$ (1).	2+1
	OR: Square: $\tan^2\theta + 2 + \cot^2\theta = 64$ (2); value $= 62$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Adjacent $= 7k$, opposite $= 24k$, hypotenuse $= \sqrt{49k^2 + 576k^2} = 25k$ (1). (ii) $\sin\theta = \frac{24}{25}$, $\cos\theta = \frac{7}{25}$ (1). (iii) $\tan\theta = \frac{24}{7}$ (1). (iv) $\sin 35^\circ = \cos 55^\circ$ and $\cot 20^\circ = \tan 70^\circ$ (1), so the value is $1 + 1 = 2$ (1).	1+1+1+2
20	LHS $= (\cos^2 A - \sin^2 A)(\cos^2 A + \sin^2 A)$ (2); $= \cos^2 A - \sin^2 A$ (1); $= \cos^2 A - (1 - \cos^2 A) = 2\cos^2 A - 1$ (2).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{6\tan A + 8}{1\tan A + 1}$ (1); $= \frac{6(\frac{4}{3}) + 8}{1(\frac{4}{3}) + 1} = \frac{48}{7}$ (2); verified with $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$ (1).	1+1+2+1

Mistakes to avoid:

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- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta = \frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.

- Writing $\sin 2A = 2\sin A$ or $\sin(A+B) = \sin A + \sin B$, which are false.
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